

1 Logic as a Science and and Art

1.1 Sciences vs. Disciplines

Before you read this introduction, chances are that you thought of logic chiefly as a *discipline*: that is, as a body of knowledge that can be acquired through study. That is not wrong, exactly; however, it is limiting. This manner of looking at logic considers it only in the abstract—that is, it ignores the *being* or the *existence* of logic as it is actually found in reality. (In the logic of concepts, we will discuss the difference between *abstract* and *concrete* concepts.) In fact, this abstract way of looking at a subject is likely how you have looked at *all* academic fields, whether that be biology, quantum physics, astronomy, psychology, economics, or any other course of study. This is a somewhat natural reduction to make, because we generally gain access to such fields through formal education in a university setting: that is, by taking classes, reading books, writing papers, doing laboratory assignments, and the like.

In reality, however, a field of study does not reduce to an abstract discipline, a set of propositions that one must acquire. Rather, acquiring knowledge in a given discipline produces a real change in the mind—or to use more precise terminology, in the *intellect*—of the person who learns it. Let me illustrate with an example. Consider the following equation:

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle$$

Readers familiar with quantum mechanics will recognize this as the time-dependent form of the Schrödinger equation.¹ For unfamiliar readers, it is doubtless completely unintelligible. In short, our human capabilities alone—in this case our *intellect*, our capacity for knowledge of things in themselves—are insufficient in order to interpret an equation of this kind. Rather, in order to understand it, we would need to undergo a stable change in our intellect. The intellect alone and unaided will not serve, but it requires a modification or ordering—or, to use the technical term, an *habitus*.² Since the intellectual *habitus* in question helps us primarily to know things, rather than to do things, we will call it a *science*.

Notice that I have not identified the *science* of quantum mechanics primarily with the *discipline*. Rather, the *science* of quantum mechanics refers to the permanent and stable modification of the intellect, the *habitus* that is acquired by studying quantum mechanics. This is not the usual, modern usage of the term *science*, but rather, it reflects how classical philosophers used it—or rather, ἐπιστήμη (*epistémē*)³ for the Greek philosophers, and *scientia* for Medieval thinkers. We will follow the classical usage of the term *science* in this introduction: an intellectual *habitus* that permits or helps us to acquire knowledge. A *scientia* is a real, existing, concrete attribute of a person—not merely an abstract body of knowledge, or discipline.

As you can probably tell from the foregoing discussion, logic is also a science, an ἐπιστήμη: not merely an abstract set of rules, but an actual intellectual *habitus* that exists (in technical terms, we say that it is *inherent*) in the logician’s very person. It will, therefore, be useful to

¹The equation describes the so-called wave function, which is used to determine the probability that a given particle is at a given location. Perhaps the most familiar use case is for describing the orbitals of electrons about an atomic nucleus.

²I am going to use the Latin term *habitus* so as to avoid confusion with the English word *habit*, which primarily refers to an involuntary “tic” or pattern of behavior. An *habitus* (pronounced with a silent initial *H*) is a permanent disposition or ordering of something to an end.

³This term, of course, is the root of the word *epistemology*, the study of science or knowledge.

explain in greater detail what a science (an ἐπιστήμη) is, and how it is different from other *habitus*⁴—and also, how to demonstrate that logic is, in fact, a science.

The Essential Characteristics of any Science

Taking its cue from Aristotle's seminal work on epistemology, the *Posterior Analytics*, Medieval thinkers have generally described science (ἐπιστήμη) as *cognitio certa per causas*:⁵ in other words, what distinguishes a science from mere opinion (Greek: δόξα, *dóxa*) is that one is able to link the individual facts known to their causes.

For example—one that is perhaps more familiar to readers than quantum mechanics—consider the example of medical science. For millennia, humans have been aware that certain diseases are contagious. Perhaps the most notorious example is during the Black Death in Europe (1347-1351): in order to avoid the spread of the disease, those who were infected, or were suspected of being infected, were often quarantined⁶—that is, isolated from the rest of society for forty days. In other words, Europeans had formed the correct *opinion* regarding the tendency of the plague to spread from host to host. However, it was not until the advent of the microscope and the development of the germ theory of disease (based largely on the work by Louis Pasteur and Robert Koch in the late nineteenth century) that the true *causes* of the plague became known—and thus effective prevention and treatment could be developed. We could say that the study of infectious disease did not truly become a *science* until the causes were discovered. Similarly, even today, a person might be aware that washing one's hands to avoid the spread of infection is a good idea, or that antibiotics are an effective treatment against bacterial diseases such as the plague, and yet not know that infection by bacteria is the *cause*. In that case, the person has the correct δόξα, or opinion, regarding these matters, but he does not possess ἐπιστήμη.

Similarly, there is the example of chemistry. For many years, scientists were aware that certain chemical species could only combine with one in other in certain proportions. For example, combining one volume of elemental hydrogen with another volume of elemental chlorine would produce, once the mixture had cooled to its original temperature, two volumes of hydrogen chloride (hydrochloric acid). However, combining two volumes of hydrogen with one volume of chlorine would produce a three-volume mixture of 1/3 hydrogen and 2/3 hydrogen chloride. The reason why the hydrogen remained a separate species was a mystery, until the atomic theory was proposed, and the existence of molecules (stable combinations of atoms) was discovered. In fact, elemental hydrogen and chlorine are both constituted by diatomic (two-atom) molecules (H₂ and Cl₂), and when they combine, they form molecules with one hydrogen and one chlorine atom (HCl). This arrangement explains why some hydrogen remains unreacted: once the chlorine is entirely consumed, it can no longer react. In other words, the phenomena observed in the laboratory were not satisfactorily explained until their *causes* were discovered.

Although the seeking of causes is what distinguishes science from opinion, Aristotle identifies three other essential characteristics of a science: namely, it must be *certain*—that is, it

⁴The plural of the Latin word *habitus* is also *habitus*.

⁵See ARISTOTLE, *Posterior Analytics* (henceforth cited as *PA*), tr. by G.R.G. MURE, I.2, 71b9-11: "We suppose ourselves to possess unqualified scientific knowledge of a thing, as opposed to knowing it in the accidental way in which the sophist knows, when we think that we know *the cause on which the fact depends*, as the cause of that fact and of no other, and, further, that the fact could not be other than it is."

⁶From the French *quarantaine* ("fortyish"), which is ultimately derived from the Latin *quadraginta*, forty.

must lead to true knowledge without fail or error⁷; it must study that which can be attributed to its object in a *necessary* way (i.e., it identifies what cannot possibly be otherwise)⁸; and it must be *universally* applicable to its object of study.⁹

Again, it can help to think of some examples. Consider the science of chemistry, which helps us to understand the properties of material entities, as regards the interactions that occur on the atomic and molecular level. It is clear that chemistry provides a great degree of *certainty*: if I mix sodium bicarbonate (H_2CO_3) with hydrochloric acid (a solution of HCl in water), I can be *sure* that table salt (sodium chloride, NaCl) and carbon dioxide (CO_2) will be produced. Similarly, one way to test to see if I have a base, such as sodium bicarbonate, is to mix a little hydrochloric acid into it; if it bubbles, I can be *certain* of it. In a similar vein, when I mix the acid into the bicarbonate, the outcome is the production of salt and carbon dioxide—and this result could not possibly be otherwise; that is, the evolution of these products is a *necessary* result. Finally, the result is *universal*, because it applies to *all* similar situations: in short, bicarbonate by nature reacts with hydrochloric acid, therefore *all* samples of bicarbonate will react with *all* batches of the acid. Another way of saying this is that chemistry studies only that which can be attributed *per se* (in and of itself) or *essentially* to the entities studied. Finally, as I mentioned above, chemistry is a science precisely because it seeks to identify the *causes* of these properties and reactions, which are to be found fundamentally in the atomic and molecular structures of the materials, especially in the arrangement and interaction of the electrons about the nuclei of the atoms.

The degree of certainty and necessity—and thus its universal applicability—can vary from science to science. Clearly, the degree of certainty and necessity that can be attained in pure mathematics is much greater than what can be obtained in a social science such as psychology or economics. Nevertheless, even in cases in which complete certainty, necessity, or universality are unattainable, a science provides useful knowledge *to the degree* that it is certain, necessary, and universal. For example, economics affirms that by and large (but not necessarily in every single case), when the price of a good or service rises, the quantity demanded falls. Although economics is unable to predict what an individual consumer will do, it does accurately account for general trends among many consumers. What a science cannot do is account for phenomena that have no essential link to their causes whatsoever; that is, science cannot account for what is *per accidens* (completely incidental). To a geneticist studying fruit flies, for example, what interests him are the laws that govern the inheritance of genetic characteristics—what the flies happen to eat for food does not interest him, as such, since it does not affect heredity.

In summary: a science or ἐπιστήμη is an intellectual *habitus* that helps the intellect to understand a given object of study. As such, it always has four essential characteristics:

1. Most fundamentally, it seeks to connect the phenomena it studies to their underlying causes. The seeking of causes is what distinguishes science (ἐπιστήμη) from opinion (δόξα), and the other three characteristics of a science all follow from this one.

⁷As Aristotle points out in the text cited above, by seeking the causes of a phenomenon, the whole purpose is to ensure that “the fact could not be other than it is” (*PA* I.2, 71b11), and as a consequence, such knowledge is free from error.

⁸See *PA* I.2, 71b14–15: “Consequently the proper object of unqualified scientific knowledge is something which cannot be other than it is.”

⁹See *PA* I.11, 77a5–9: “So demonstration does not necessarily imply the being of Forms nor a One beside a Many, but it does necessarily imply the possibility of truly predicating one of many; since without this possibility we cannot save the universal, and if the universal goes, the middle term goes with it, and so demonstration becomes impossible. We conclude, then, that there must be a single identical term unequivocally predicable of a number of individuals.”

2. It provides certain knowledge of its object—that is, it enables the one who possesses the science to know without error.
3. It accounts for phenomena that cannot be different from the way they are, or cannot come about in a different way. That is, it accounts for phenomena that are linked necessarily (*per se*), and not merely incidentally (*per accidens*), to the object of study.
4. Finally, it applies universally to the object of study: there is no science, as such, of singulars and particulars.

A couple of things follow from this. First of all, in order to determine whether a given intellectual *habitus* is a science, it is sufficient to show that it has these four characteristics. We will revisit this topic below. Second, it is not possible for science or ἐπιστήμη, in the strict sense of the term, to be false; that is, false information is the result, not of science, but of erroneous opinion (δόξα).

Speculative vs. Practical Sciences

We are not, however quite through with our examination of science: as it turns out, sciences come in different kinds, depending on their object of study and method. By now, you may have intuited that by defining science as the ancient and Medieval philosophers did—as *scientia* or ἐπιστήμη, that is, as an intellectual *habitus* whose purpose is to know the properties of a given object of study according to its causes—we are including under the banner “science” many more fields than are customarily placed there. By this definition, “science” is not limited to empirical fields such as biology, chemistry, or physics, but rather includes areas of study as diverse as mathematics, metaphysics, theology, and ethics. It is sufficient that each field have a specific and unique object of study (the *genus subiectum*) and appropriate methods for it.

Sciences can be categorized according to their finality—a criterion that is particularly useful for us. In the modern parlance, we speak of *pure sciences* such as biology, chemistry, and physics, and contrast them to *applied sciences* such as medical science and engineering. This division can be generalized to apply to all sciences (in the classical sense, as we have defined the term here): that is, some sciences seek knowledge of their object for its own sake, whereas other sciences seek the applications of the knowledge gained from them to some action. In the classical terminology, the former are called *speculative sciences*, whereas the latter are called *practical sciences*.

Again, it can be helpful to look at some examples. Consider the sciences listed in Table 1. What the speculative sciences listed in the left-hand column have in common is that they all seek knowledge of the object they study for its own sake. For example, a biologist can study the behavior of nesting robins without having any practical purpose whatsoever. This or that scientist may in fact have ulterior designs (he may be thinking of starting a business raising robins for their eggs), but that ulterior motive (or lack of one) does not change the character of his work *as a biologist*. The same can be said for chemistry, physics, mathematics, and metaphysics¹⁰.

On the other hand, the practical sciences listed in the right-hand column all have a specific kind of operation (that is, a kind of action), in mind. What they produce is still a kind of *knowledge*, but in contrast to speculative sciences, for practical sciences, knowledge is not an end in itself—it is a means to action. For example, studying physics by itself will not

¹⁰In fact, metaphysics, the study of being (that is, of *ens*, that which is) as being, or being *qua* being, is rightly considered the most speculative of all sciences, since it has no direct practical application.

Table 1: A few examples of speculative and practical sciences. As much as possible, each practical science is placed next to the speculative science it is derived from.

Speculative	Practical
Metaphysics	—
—	Ethics
Biology	Medical science
Chemistry	Chemical engineering
Physics	Mechanical engineering
Mathematics	—

be enough to tell you how to build a bridge; for that, you would need to study mechanical engineering. You need to learn how to apply the general principles discovered in physics to the more concrete problem of how to build structures. Similarly, studying biology—even the human biology—by itself is insufficient to discover how to cure disease; that is the task of medical science. As you can probably tell from these examples, many practical sciences can be paired with a speculative science from which they are derived. In a similar vein—although it may not be obvious at first—ethics, the study of human actions (actions that involve the free choices made by the will) in order to discern whether they are right or wrong—is also in the column of practical sciences. This is so, because the entire purpose of ethics is to direct our actions: we wish to discover which behaviors are compatible with human happiness, and which ones are not—in other words, the knowledge obtained through ethics is a means to the end of behaving well.

Is Logic a Science?

It is now time to turn our attention once more to our topic—logic—and to apply the principles we have been discussing. As we saw above, logic is clearly an academic discipline and field of study (at least when taken in the abstract), but the question now arises: is logic a *science* like biology, chemistry, and mathematics? In order to answer this question, we will need to apply our definition: a science is a *cognitio certa per causas*.

Is logic a *cognitio*—that is, an *habitus*, or stable disposition, of the intellect? In other words, in the absence of logic, would the intellect be impeded from acquiring certain types of knowledge? We will see this more clearly when we give the definition of logic below, but even based on our intuitive understanding of logic, we can confidently answer “yes”: in fact, in the complete absence of logic, we would be unable to acquire any knowledge *at all*—and a systematic study of logic makes our thinking easier, more ordered, and less prone to error.

Does logic provide sure and certain knowledge by linking what it studies to its causes? As we will see, the answer is again “yes.” Logic seeks the *causes of knowledge*: for example, through logic, I can be sure that conclusion *C* follows from premises *A* and *B*. This type of causality is different from that sought, say, in the sciences I listed above, all of which seek the *causes of being*, rather the causes of knowledge. That is, the *logical* causality proper to logic is distinct from the *ontological* causality proper to other sciences—but it is still causality.

Finally, what about the other essential characteristics of a science? Does logic provide *certain* knowledge—that is, knowledge without error? Does it supply knowledge that follows necessarily from the nature of its object of study? Does the knowledge it furnishes apply

universally to its object? Again, the answer to all three questions is in the affirmative. As regards certainty, the very purpose of logic is to facilitate our reasoning ability and make it less prone to error. As regards necessity, again, logic provides the tools we need to see that the conclusions we reach are the correct ones, and could not be otherwise. As regards universality, it is clear that logic applies to *all* of the works of our intellect, without exception.

In fact, we should carefully define our terms, here, because there are two distinct entities without which knowledge is simply impossible: there is our intellect, which is the faculty that permits us to make acts of knowledge—knowledge that captures the very essence and being of things; and there is the intellectual *habitus* that we are discussing here. For this reason, intellect, which is sometime called our *reason* (in Latin: *ratio*; in Greek: λόγος or *lógos*)¹¹, can be called a kind of “natural logic.” However, logic, in the strict sense of the term, refers to an *habitus*, an acquired ability that *helps* the intellect in its operation. However, it should be noted that this logic-as-science can be acquired in an imperfect way (that is, the intuitive grasp of logic that we acquire from the mere use of our reason), or else in a “perfect” or systematic way—which is what this course is attempting to do.

We must, therefore, conclude that logic is indeed a science.

Logica utens vs. logica docens

Based on our previous discussion, the another question arises: is logic a *speculative* science or a *practical* science? As we saw, the purpose of a speculative science is knowledge for its own sake. Logic certainly seems to fit the bill. St. Thomas Aquinas in his *Summa theologiae* describes the purpose of logic in this way:

[A] certain art is required, one that can direct the very act of reason, through which man, if you will, in that same act of reason may proceed in an orderly manner, more easily, and without error, and this art is logic, that is, the rational science.¹²

Thus, the purpose of logic is to order the individual actions of our intellect in such a way that the acquisition of knowledge is easier and less error-prone. When someone has acquired the science of logic, it will be much easier for him to distinguish truth from falsehood; in fact, we could summarize Aquinas’ description here as saying that logic is our instrument for discovering true knowledge, and ensuring that it is *true*.¹³ As such, we can categorize logic as a speculative science, albeit a unique one, because (as we will see) its ultimate goal is not knowledge of its object—the works of reason—but rather knowledge of the objects of all the *other* sciences. That is, we do not study logic in order to study the works of reason for their own sake; rather, we study the works of reason in order to *use* them in biology, mathematics, physics, engineering, and every other science (including logic itself)—indeed in every act of the intellect. We can summarize this idea by saying that logic is *instrumentally speculative*: it

¹¹In the strict sense, our *reason* refers to our capacity for acquiring new knowledge starting from knowledge that we have already acquired: that is our ability to infer or draw conclusions, or, to use the technical term, our *ratiocinium*. However, when taken broadly, the term *reason* is synonymous with the intellect.

¹²THOMAS AQUINAS, *Expositio libri Posteriorum Analyticorum* (henceforth cited as *EPA*), proemium 1-2: “[A]rs quaedam necessaria est, quae sit directiva ipsius actus rationis, per quam scilicet homo in ipso actu rationis ordinate, facilius et sine errore procedat. Et haec ars est Logica, idest rationalis scientia” (my translation).

¹³A reminder that, for the ancients, *knowledge* (*scientia*, ἐπιστήμη) as such is always true. There can be false *opinion* (δόξα), but not false *knowledge*; we will follow that convention in this course. (I will also mention that the ancients used the same term *scientia* or ἐπιστήμη to refer both to the *habitus*—that is, the habitual knowledge that is not necessarily in act—and to the individual *acts* of knowledge that flow from the *habitus*.)

certainly seeks knowledge for its own sake, but it does so strictly *as a means or instrument* of obtaining that knowledge.

A new question immediately arises: if logic is a means to obtaining knowledge, and obtaining knowledge is an action, doesn't that make logic a practical science? It does indeed. What this means is that—somewhat surprisingly—"speculative" and "practical" are not mutually exclusive: a science can be both.

How is this possible? It goes back to what we touched on earlier: a science, being an intellectual *habitus*, makes it possible for the intellect to do certain actions that it was unable to do before. An *habitus* is only required when there is a special difficulty in acquiring the knowledge in question: for instance (referring to the example above), I only need to learn (that is, acquire habitual knowledge of) quantum physics if I need to understand things like the Schrödinger equation. As we discussed above, many speculative ("pure") sciences can be paired with a practical ("applied") science—for instance, mechanics (physics) with mechanical engineering, or human biology with medical science. Why is a distinct applied science necessary? In short, because the application is not always obvious: mechanics, for example, tells me how to calculate the forces acting on a rigid body; it does not tell how to apply this knowledge to building a bridge. Similarly, human biology tells me that cancers are caused by genetic mutations that accumulate in the DNA of the body's cells; it does not tell me the best *treatment* for removing that cancer. In sum: when the application of speculative knowledge poses a particular difficulty, a distinct *habitus* (in this case, a distinct *science*) is required in order to make that application.

With logic, however, the situation is a bit different. Since logic studies the very works of our reason and (as we will see) the proper construction of relations between them, merely studying the object of logic immediately hones our ability to reason. We will return to this idea below.

In any case, it is clear that, in addition to being a speculative science, logic is also a practical science: although logic is speculative insofar as its end goal is knowledge, its manner (or *mode*) of achieving that goal is practical: said in other terms, logic teaches us *how to know*. For this reason, logic is said to be a *modally practical* science.

Is Logic Also an Art (τέχνη)?

Attentive readers will note that Aquinas, when describing the goals of logic, does a curious thing: he calls it an *art* (*ars*; in Greek: τέχνη or *téchne*). That is surprising, because we have just spent the last few pages demonstrating that logic is a *science*. It may also surprise readers that Aquinas calls it an "art" along with painting, music, sculpture, and theater. In order to understand this, it is best to take a moment to understand what Aristotle and Aquinas meant by *art* or τέχνη.¹⁴

For Aquinas, an art could refer to any intellectual *habitus* that results immediately in action: in the classical formulation, *recta ratio factibilium*, right reason applied to things that are made. In other words, it is a use of reason or the intellect (*ratio*) that is properly ordered (*recta* or "right") to the accomplishment of determined operations or actions (*factibilium*).

Again, some examples are helpful: suppose you are given a piece of marble, and instructions to carve it into a statue of David. For the vast majority of readers, the result would most likely not look anything like Michelangelo's famous statue of David. As we saw before with the sciences, in order for the sculptor to be able to set chisel to stone and produce a work of beauty, he must first acquire an *habitus*—in this case the art of sculpture. I think it is clear

¹⁴From the Greek word τέχνη, we derive a number of English words, including *technique* and *technology*.

that the same reasoning applies to a violinist, an opera singer, a painter, and a photographer. However, for Aquinas, *art* is not limited to what we would call the “fine arts”; it applies to any *habitus* that is necessary for skilled action. Therefore, surgeons, architects, athletes, lawyers, carpenters, writers, orators, accountants, and entrepreneurs are all “artists” by this definition.

Notice that arts can easily be distinguished by the kind of action they produce. Some arts, called the *mechanical arts*, produce something that is external: for example, the art of painting results in physical artwork such as a fresco or canvass, and carpentry results in woodwork such as tables and chairs. On the other hand, the *liberal arts* produce something that is internal to the intellect: for instance, the art of practicing law does not leave a physical work, but rather produces intangibles such as legal precedents, decisions, motions, and verdicts.

As with the sciences, arts have certain characteristics in common. Let us take, for example, the art of carpentry. A carpenter takes a raw material—in this case, wood—and fashions it into finished products, such as tables and cabinets. In order to do so, he applies the techniques that he has learned: how to cut, trim, and smooth out the wood, putting the pieces together and gluing them in the right way, adding the right kind of finish, and so on.

We can generalize this idea to every art: the practitioner of an art takes some indeterminate entity—its “raw material”—something that *can* be fashioned as desired, but *need* not be. He transforms it using means that are determined and universal, so as to produce the desired product. That is, we could say that, in general, an artist’s task is to take an indeterminate “matter” and remove its indetermination. For instance, when Michelangelo Buonarotti was given blocks of marble, the number of possible statues he could make from them was infinite; it was fully *indeterminate*. However, the more he worked on them, the closer they became to being fully *determinate*: say, a Pietà.

In fact, any *habitus* with these two characteristics can be classified as an *art* or τέχνη. With the mechanical arts it is easy to see, but the same holds for all of the liberal arts: a defense lawyer must take all of the facts of the case and his knowledge of the laws to formulate a cogent defense for his client; an orator must take the infinite possibilities of a language and formulate a beautiful speech; a composer must take tones, rhythms, dynamics, and instrumentation, and form a musical composition.

So, what about logic? Could logic be classified as one of the arts? To summarize what we saw just now, an *habitus* is an art if it fulfills the following characteristics:

1. It takes an indeterminate matter that does not infallibly attain its fully determined form.
2. It transforms that matter into a finished product using determinate and universal means.

Does logic qualify? (1) Regarding the first condition, as we shall see, the task of logic is to take the works of our reason (concepts, enunciations, and arguments) and to construct relations between them according to very strict and determined rules. Even though our reason is ordered to the truth, there is nothing preventing us from getting the relations between them incorrect: that is, we can fall into error and falsehood. Therefore, these works of reason constitute the *indeterminate matter* of logic. (2) Regarding the second condition, among all of the possible relations among the works of our reason, logic constructs—using *universal and determined rules*—the *correct* relations, the ones that will produce *true* knowledge. It follows that logic is indeed an art: a liberal art, evidently, as it does not produce a external product.

How Logic Manages to Be All Three at Once

I just want to finish this section with a few more observations about the different *habitus* that we have been discussing. Ordinarily, speculative sciences, practical sciences, and arts can be clearly distinguished from one another, even when their subject matter is closely related. Continuing the example I mentioned above, human biology is the speculative science at the root of all medical matters; medical science is the practical science that applies the knowledge of human biology to the treatment of disease; and medicine is the art that applies medical science to concrete acts of treatment. Three distinct *habitus* are at play here, because neither of the two applications is trivial. A surgeon, for example, can hardly expect to be sufficiently trained simply by acquiring the speculative science of human biology. In fact, not even detailed knowledge of medical science is enough: he can spend years learning *how* surgeries are done, and yet be completely unprepared for performing an actual surgery. In short, he does not become a surgeon without first acquiring the art of surgery—which in this case implies not only speculative and practical knowledge, but also a training in the coordination of the hands. In other words, what renders the three *habitus* distinct from one another is that there is, if you will, a considerable gap to close between the speculative science and the practical science, and between the practical science and the art.

With logic, however, we use our intellect to study works produced by that same intellect. Therefore, the distance that is present in other fields is absent in logic. We have already noted above that the speculative aspect of logic, as a science, cannot be really distinguished from its practical aspect. It is also clear from our discussion of arts that the *art* of logic cannot be really distinguished from the *science* of logic. In other words, formal, scientific instruction in logic—what is traditionally called *logica docens*¹⁵—being the study of an object that is intrinsic to the very intellect that studies it, results immediately in an improvement in the exercise of logic—what is traditionally called *logica utens*¹⁶. Later on in the course, you will learn to distinguish and relate different kinds of concepts, how to relate enunciations to one another, and how to form and analyze syllogisms. You will be able to use those skills immediately; there is no need for an *additional* science, nor for a distinct *art*, in order to apply them.

In sum: we conclude from the foregoing discussion that logic is simultaneously an instrumentally speculative science, a modally practical science, and an art, whose object is the works of our reason. What remains to discuss in greater detail, therefore, is the *object* of logic, which we will do after a brief discussion of some key concepts in philosophical anthropology that are helpful for understanding what we have just gone over.

¹⁵From the Latin *doceo*, “to teach.”

¹⁶From the Latin *utor*, “to use or employ.”